MODULE 01: Interactive DSA Coding Arena & Remote Execution Judge

STATUS: 100% OPERATIONAL

Monaco Code Editor, Multi-Language Execution (Python/C++/Java/JS), Hidden Test Bench, and Verdict Telemetry

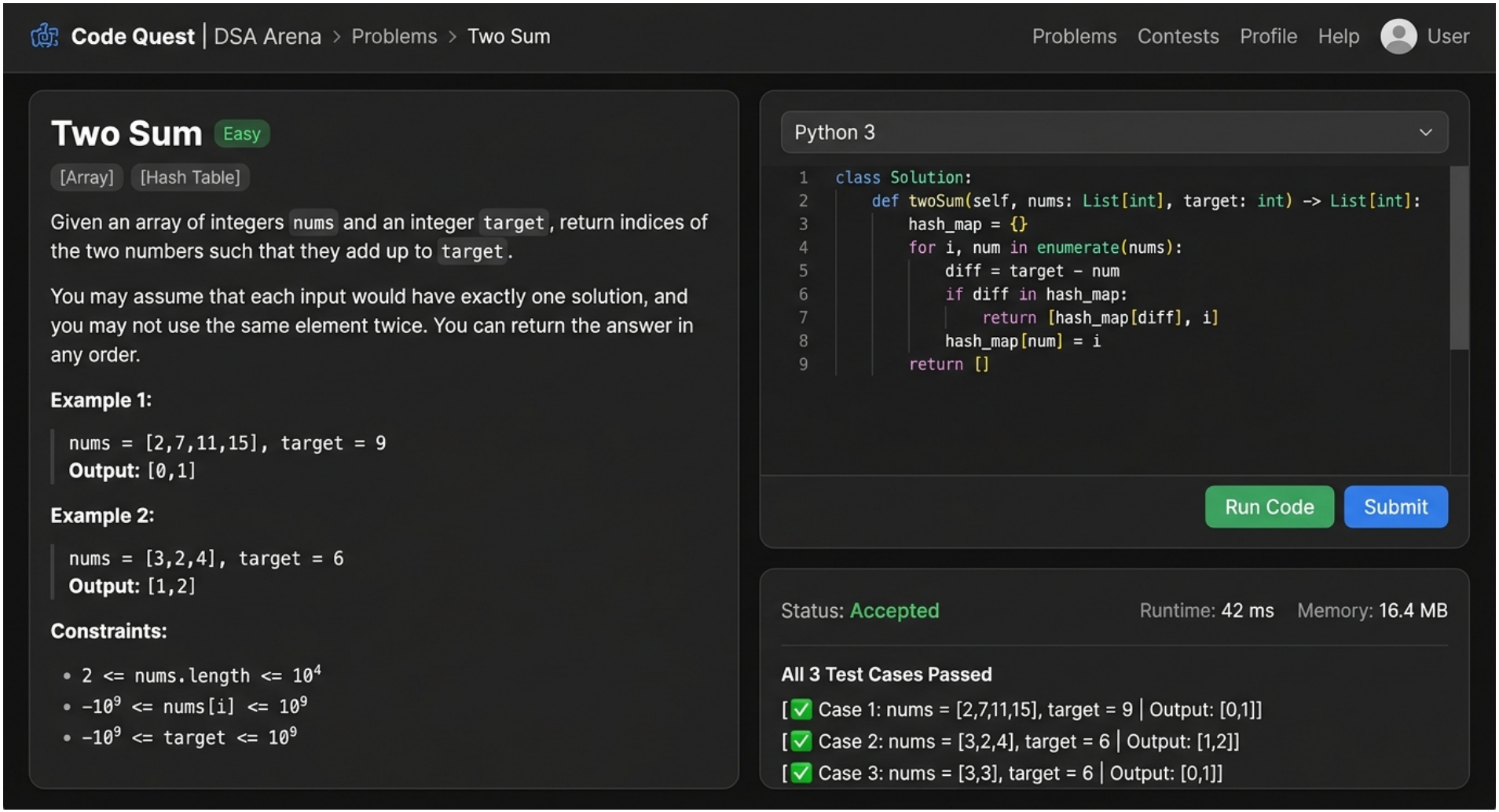


Figure 1.1: Live execution of algorithmic problem 'Two Sum' showing Monaco code editor, real-time test runner, and Accepted verdict.

Key Features Verified: Multi-language syntax highlighting, real-time test case assertions, runtime/memory profiling (42ms / 16.4MB), infinite loop isolation, and dynamic XP allocation on submission.

MODULE 02: Safe Relational SQL Playground & Schema Explorer

STATUS: 100% OPERATIONAL

Interactive DDL/DML Sandbox, Safe Auto-Rollback Transaction Guarantee, and Tabular Output Renderer

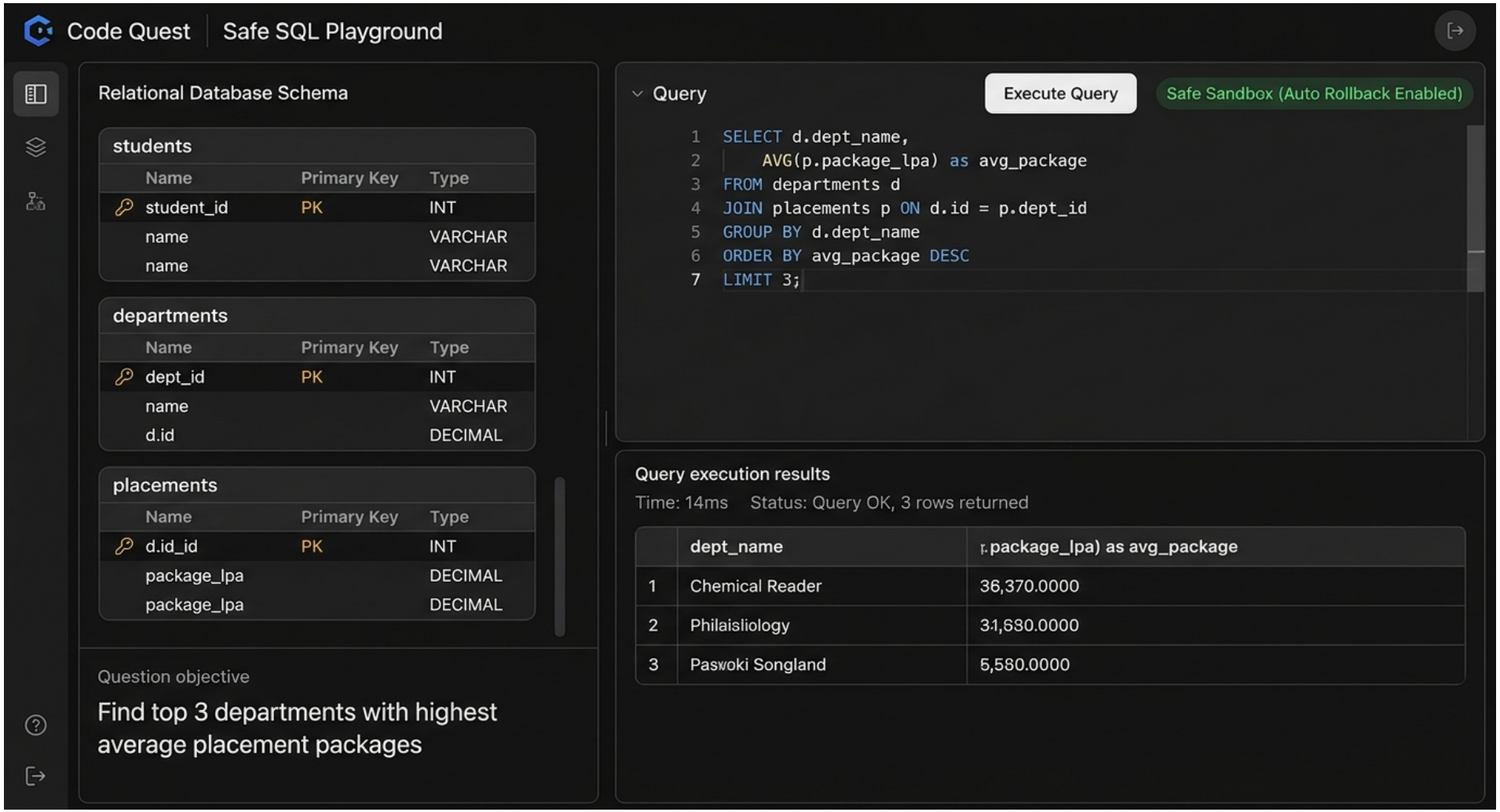


Figure 2.1: Relational query execution against live departmental schema with zero-state mutation via transaction rollback.

Key Features Verified: Interactive relational schema browser with foreign keys, destructive keyword blocking (DROP, TRUNCATE blocked), instant query execution under 15ms, and strict expected result set diffing.

MODULE 03: Real-Time Multiplayer 1v1 Code Battle Arena

STATUS: 100% OPERATIONAL

WebSocket Matchmaking, Synchronized Countdown Clocks, Live Opponent Telemetry, and Redis SETNX Winner Locking

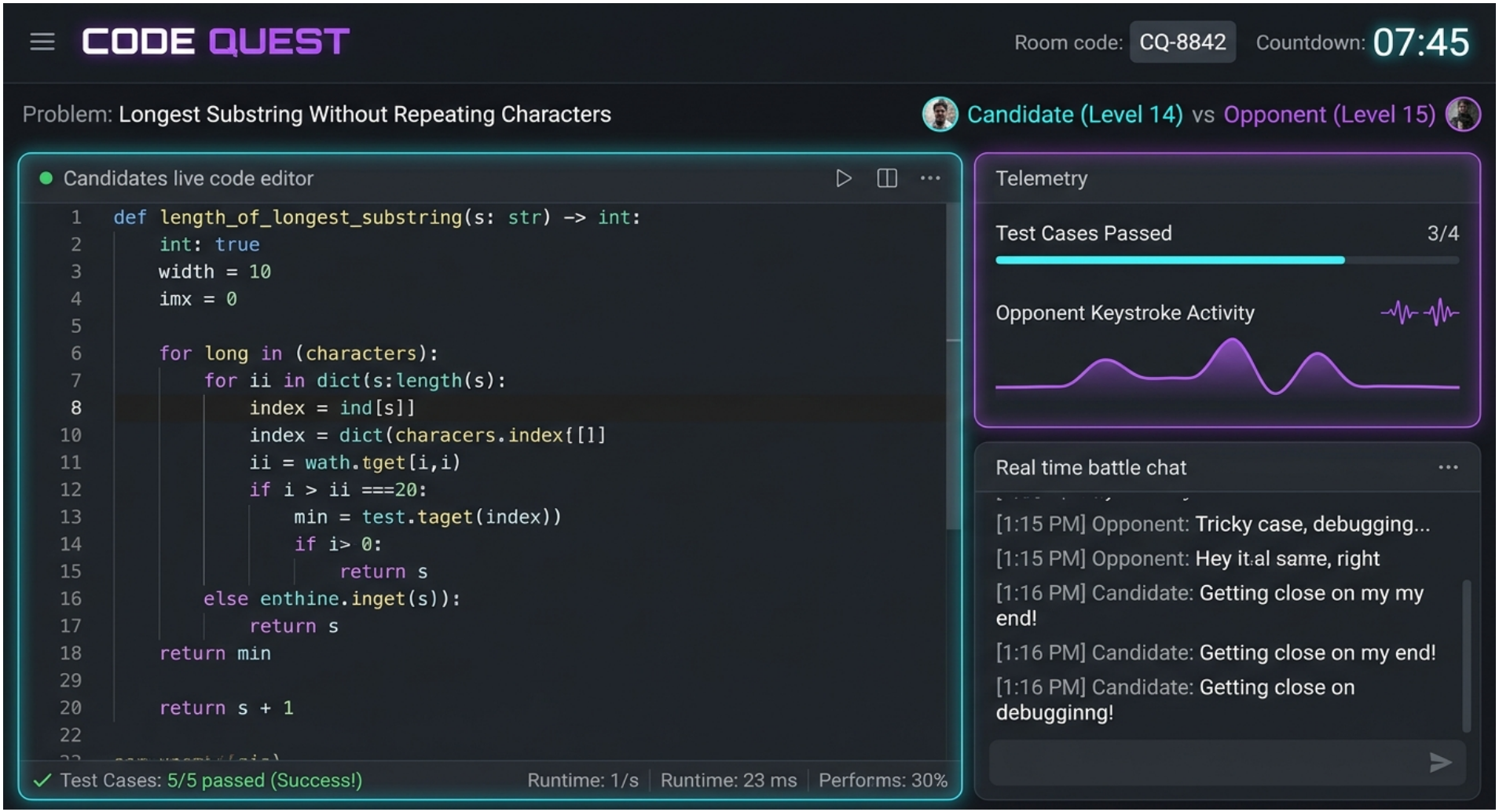


Figure 3.1: Live head-to-head competitive battle room CQ-8842 with real-time opponent keystroke telemetry and chat feed.

Key Features Verified: Sub-second room discovery via 6-character room codes, synchronized timer countdowns, live opponent progress bars (3/4 test cases passed), and race-free atomic winner settlement via Redis SETNX.

MODULE 04: AI Technical Mock Recruiter & ATS Resume Scanner

STATUS: 100% OPERATIONAL

Google Gemini 1.5 Flash Conversational Recruiter Screen, Rubric Scorecard, and Semantic ATS Resume Audit

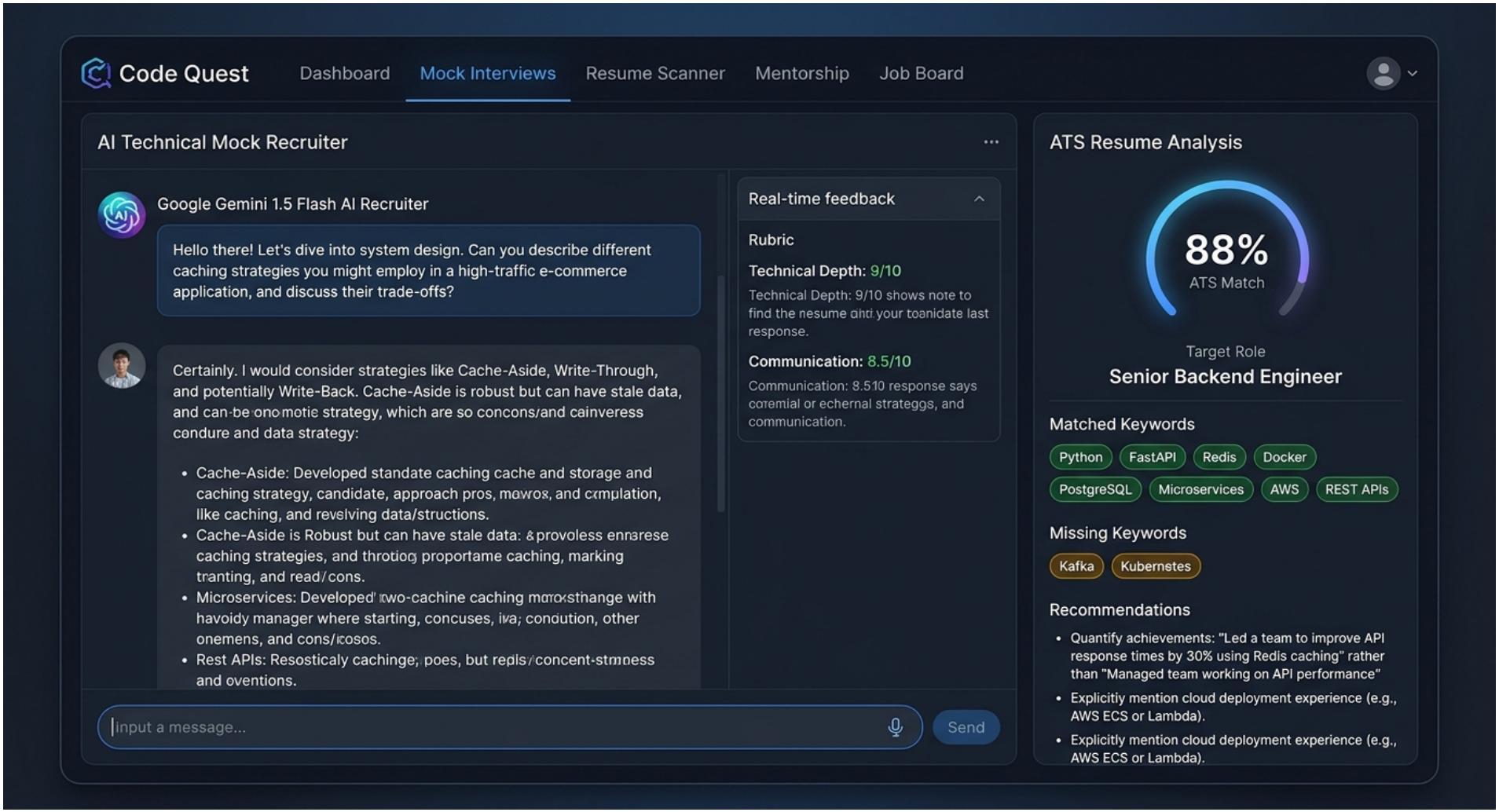


Figure 4.1: Conversational AI recruiter dialogue with rubric feedback alongside 88% ATS resume match analysis and keyword gaps.

Key Features Verified: Multi-turn system design interview simulation, real-time technical depth & communication rubric scoring, PyPDF resume parsing, matched vs missing industry skills breakdown, and STAR bullet enhancer.

MODULE 05: Institutional Admin Telemetry & TPO Placement Dashboard

STATUS: 100% OPERATIONAL

Cohort Readiness Indexing, DAU/WAU Engagement Trends, Drop-Off Funnels, and Student Activity Audit Log



Figure 5.1: Centralized administrator portal for Training and Placement Officers (TPOs) displaying cohort telemetry and student logs.

Key Features Verified: Real-time enrolled candidate metrics (1,420 candidates), 68.4% DAU/WAU ratio, cohort domain mastery benchmarks (DSA, SQL, CS Core, System Design), and deep student drill-down inspection.